

Sharp Solved Families and Constant-Factor Paths in Polynomial Lemniscates

*Unconditional parent progress, exact obstructions, and frontier mechanisms for Erdős
Problem #1041*

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ABSTRACT

Let $f(z) = \prod_i (z - z_i)$ be monic, with every root in the open unit disc. Erdős, Herzog, and Piranian asked whether two roots can always be joined inside $\{|f| < 1\}$ by a curve of length less than 2. The problem remains open. Our strongest theorem applying to arbitrary degree and arbitrary root geometry is an unconditional constant-factor version. Put

$$\mu = \min_{f'(c)=0} |f(c)|, \quad \rho = \mu^{1/n}.$$

Two zero occurrences are joined inside $\{|f| \leq 2\mu\}$ by a path of length at most $(71/10)\rho$. If the roots lie in the unit disc and $\mu \leq 1/2$, the construction lies in $\{|f| < 1\}$ and has length at most 5.7. It proves the target constant 2 whenever the first-merge component has at least 17 roots, and also under explicit arity-dependent component-capacity defect inequalities. These are ordinary analytic theorems, not Lean formalisations; no literature-priority claim is made.

We also prove the conjectured path bound for three classes of polynomials. For collinear roots the result is all-degree and sharp: one adjacent-root segment works with the best Chebyshev constant. The other complete classes are translated cubic quotient fibres $P((z-h)^q)$ and primitive sparse quintics $z^5 + az^4 + bz + c$. Lean checks their principal selectors and extremal inequalities; the finite-fibre, path-assembly, and affine-transport arguments are proved in the paper.

At every non-root critical point c we prove the sharp metric selector: if $n \geq 2$ and

$$r^n = \prod_{k=1}^n |c - z_k|,$$

then two distinct roots satisfy

$$|c - z_i| + |c - z_j| \leq 2r.$$

Lean checks this theorem. It also checks two counterexamples showing why the bound does not immediately give the required path: the nearest critical spoke can leave the unit lemniscate, and some cubics admit no valid pair of straight spokes through the critical point.

**Authorship and AI use.* Will Cook built and directed the research infrastructure and reviewed the claims when he could. AI agents did most of the research and drafting. Cook did not independently verify every claim.

Finally, we identify two specific obstacles to the remaining argument. The merge-tree life-time is governed by the strictly convex function $\Phi(x) = \int_0^x (\log(\coth t))^{-1} dt$, for which $\Phi(x)/x \rightarrow 0$ as $x \rightarrow 0$; no positive uniform linear conversion from attachment age to life-time is possible. Separately, a printed spanning-tree argument fails both at a four-pronged saddle and on an exact Cassini example. The unresolved work is therefore to control the relevant merge ratios and to construct a compatible planar path with total length below 2.

1 The problem

Problem 1.1 (Erdős #1041). Let $f(z) = \prod_{i=1}^n (z - z_i)$ be monic with $z_i \in \mathbb{D}$, the open unit disc. Show that two of the roots can be joined by a curve of length less than 2 lying in the open lemniscate $E = \{z \in \mathbb{C} : |f(z)| < 1\}$.

Numbering and current status follow Bloom’s Erdős problem catalogue [1]. The problem is open. The original source is Problem 5 on printed p. 139 of Erdős–Herzog–Piranián [2, p. 139]; the preceding paragraph records the known input that one component of the lemniscate contains at least two zeros.

Two recent manuscripts are relevant. The 48-page manuscript posted by shtuka on 24 March 2026 [3, Theorem 1, p. 1] claimed the unrestricted statement. Its Proposition 12 (p. 16, with proof continuing through p. 30) supplies the spanning-tree decomposition used in its final proof. That manuscript has been withdrawn by its author, who wrote on the problem’s discussion page on 26 March 2026 that the statement of Proposition 12, and not only its proof, is incorrect, after Tao identified an unjustified invocation of its Lemma 8 the previous day. The file has not been revised since 24 March 2026. Section 9 gives an exact Cassini counterexample to Proposition 12’s metric conclusion, reached independently; the same witness refutes Proposition 7 of an unattributed local revision, and the local three-ended saddle construction fails separately. A second general-degree attempt circulated on 22 April 2026 and was broken on 27 April 2026. No general-degree manuscript stands, and the problem page records the problem as open with no accepted proof claim. Pendyala’s independent June 2026 preprint [4, Thm. 1, p. 1] proves the degree-four case. The quartic theorem does not close the problem. An adjacent June 2026 paper by Pendyala [5, Thm. 1.2, p. 2; introductory discussion, pp. 1–2] proves $c\sqrt{\log n} \leq S(n) \leq \pi n$ for all sufficiently large n , where $S(n)$ measures the shortest path from 0 to $\partial\mathbb{D}$ in the normalized set $\{|z| \leq 1, |f(z)| \leq 1\}$. This is a different statement from the root-to-root problem here: it supplies important adjacent lemniscate-path context, but it is not evidence that Erdős #1041 is solved or even a partial result for its root-pair conclusion.

The two primary comparisons are pinned to downloaded PDFs. The Erdős–Herzog–Piranián source and digest are

annexes/erdos-herzog-piranián-1958-metric-properties-of-polynomials/source.pdf
SHA-256: bff39876de5e152b5a0d2b622eb6f9ca0c747d401ceea68146cfa2b2b5f28ce9.

Its exact locator is printed p. 139 / PDF p. 15 / extracted lines 594–602, where the unit-disc setting, the two-zero component, and Problem 5 appear. It supports the historical problem statement only, not the local length–area estimate or the all-degree constant-factor result. Pendyala’s downloaded degree-four source and digest are

annexes/arxiv-2606-24875-pendyala-degree-four-lemniscate-path/source.pdf
 SHA-256: 337db0cc046023b1429d9131720d199843766629cd2ad8bf8b04a40168d7bcd8.

Its abstract and Theorem 1 are on PDF p. 1, the four-point radial lemma is on PDF pp. 1–2, and the theorem proof runs through p. 3. That proof’s minimum-enclosing-disk/four-point mechanism is a genuine degree-four result, but it does not extend the all-degree claims or validate the Newton-flow decomposition below.

The adjacent shortest-path comparison is likewise pinned to its downloaded primary source. Pendyala’s 34-page arXiv:2606.19178v1 PDF was downloaded from <https://arxiv.org/pdf/2606.19178v1>; the arXiv API record is https://export.arxiv.org/api/query?id_list=2606.19178. Its digest is

SHA-256: d7902a8bc37d2aad6dea52dffbbbaaf7ff23954ece5a0c4cbab108f58df4e6df.

The abstract (printed p. 1, extracted lines 12–32) and Theorem 1.2 (printed pp. 3–4, extracted lines 97–113) support the bounds $c\sqrt{\log n} \leq S(n) \leq \pi n$. The introduction’s distinction from the root-to-root problem is on printed pp. 2–3 (extracted lines 51–78); the reciprocal-sweep and alternating-maze mechanisms are described in extracted lines 79–96. These locators support only the neighboring marked-point result: they do not support a root-to-root path, a degree-free bound, or a transfer to the first-merge problem.

The object we work with is not the lemniscate directly but the flow that foliates it.

Status. The problem treated here is open, and this note does not close it. Every statement below marked as checked is a proposition that the pinned Lean kernel accepts from the sources this note links to, with no sorry, no added axiom, and no unchecked evaluation. That is a claim about the formal statement, not about its mathematical interest, its novelty, or the original problem. The unresolved obligations are named exactly, in their own section, and none of the finite computations, reductions, or no-go results here removes one of them.

Companion system context. The [claim and trust boundary](#), [cold-clone route to proof authority](#), and [public contribution protocol](#) are described in sibling papers. Those descriptions do not change the mathematical status of this note.

Main results at a glance

The problem remains open. The table is ordered by mathematical consequence: sharp solved families and unconditional progress on the parent geometry come before auxiliary selectors and mechanism eliminations. Section 16 gives the complete evidence and boundary ledger.

Result	What is established
Sharp collinear family	The conjecture holds in every degree for collinear roots, with the sharp Chebyshev constant. CHECKED KERNELS; ORDINARY AFFINE TRANSPORT.
Global constant-factor theorem	For arbitrary monic f , two zero occurrences are joined in $\{ f \leq 2\mu\}$ by a path of length at most $(71/10)\mu^{1/n}$. For unit-disc roots and $\mu \leq 1/2$, the path lies in $\{ f < 1\}$ and has length at most 5.7. ORDINARY EXACT PROOF.
Parent-theorem regimes	The target constant 2 holds when $\mu \leq 1/2$ and the first-merge component has at least 17 roots, or under an explicit arity-dependent component-capacity defect bound. ORDINARY EXACT PROOF.
Critical-value separation regime	If a simple critical point c with $v = f(c) \neq 0$ has every other critical point d at normalised distance $ 1 - f(d)/v \geq S > 1$, two roots are joined inside $\{ f \leq v \}$ by a path of length below $2 v ^{1/n}$ whenever $(1 + S)^{2/n} \log(S/(S - 1)) < 1$. The exact regimes are $S = 4$ for $n \geq 3$, $S = 3$ for $n \geq 4$, and $S = 2$ for $n \geq 6$, and each cutoff is the first degree at which its inequality holds. ORDINARY EXACT PROOF; LEAN CHECKS THE DEGREE MONOTONICITY, THE THREE THRESHOLDS, AND THE SQUARED-LENGTH CONSUMER.
Critical-spectrum closure	At a simple minimising critical point, enough separation from the next critical-value modulus gives a join inside the sharper level K_μ and an explicit degree-dependent criterion for the target constant 2. ORDINARY EXACT PROOF.
Primitive sparse quintics	Two exact tail-energy selectors give the conjectured path for the open-disc sparse-quintic family. CHECKED SELECTORS; ORDINARY PATH ASSEMBLY.
Translated cubic quotient fibres	The conjectured path exists for every $f(z) = P((z - h)^q)$ with P monic cubic and $q \geq 2$. CHECKED KERNEL; ORDINARY FINITE-FIBRE ASSEMBLY.
All-degree proximity theorem	At every non-root critical point, two roots have total critical-point distance at most twice the geometric mean of all root distances. LEAN CHECKED.
Straight-path obstructions	Exact quintic and cubic families rule out the unique-nearest spoke and every pair of straight spokes as universal constructions. LEAN CHECKED.
Current boundary	The general problem is still open. A counterexample in the $\mu \leq 1/2$ branch must have first-merge arity at most 16 and lie above the corresponding capacity-defect cutoff; the high-critical branch also remains.

2 An unconditional all-degree constant-factor path theorem

The parent problem asks simultaneously for containment in the open unit sublevel and the sharp length 2. The following theorem keeps the same root-to-root path object for arbitrary geometry and arbitrary degree, while relaxing both constants in a controlled scale-invariant way. For a monic degree- n polynomial put

$$K_t = \{z : |f(z)| \leq t\}, \quad \mu = \min_{f'(c)=0} |f(c)|, \quad \rho = \mu^{1/n}.$$

Theorem 2.1 (unconditional constant-factor path). *For every monic polynomial f of degree $n \geq 2$, two zero occurrences are joined by a possibly degenerate path of length at most*

$$\frac{71}{10} \rho$$

inside $K_{2\mu}$. If f is squarefree, their locations are distinct. If every root lies in the open unit disc and $\mu \leq 1/2$, the construction may be chosen inside $\{|f| < 1\}$ with length at most 5.7.

The constant is the rationally certified specialization of a two-parameter bound. If the selected working component contains $k \geq 2$ roots, then for every $r \in (0, 1)$ and $\lambda > 1$ the proof constructs a path in $K_{\lambda\mu}$ whose length is at most

$$\sqrt{\frac{2}{k}} \left(\frac{\sqrt{2}r}{(1-r)^2} + \lambda^{1/n} \left(\sqrt{\log(\lambda/r)} + \frac{\pi}{\sqrt{\log \lambda}} \right) \right) \rho. \quad (\text{CF})$$

For $n \geq 3$, the choice $\lambda = 2$, $r = 3/20$ makes the bracket less than $71/10$; degree two has the exact root-segment bound 2ρ .

Proof architecture. Choose a critical point at level μ and a component at a regular level $t \in [\mu, \lambda\mu]$ that contains the corresponding first merge. If a regular component C' at level σ contains k' roots, winding and Cauchy–Schwarz give the local perimeter estimate

$$\mathcal{H}^1(\partial C')^2 \leq 2\pi k' \sigma \frac{d}{d\sigma} \text{Area}(K_\sigma \cap C'). \quad (\text{CF1})$$

Pólya's area–capacity inequality $\text{Area}(K_T) \leq \pi T^{2/n}$ then selects t with a degree-free boundary-length bound.

For an argument avoiding the finitely many critical-value rays, lift the radial value segment from each root to ∂C_t . Split each lift at level $r\mu$. Koebe distortion controls the low pieces. The exact precritical conformal-radius energy inequality

$$\sum_i \frac{\mu^2}{|f'(z_i)|^2} \leq \rho^2 \quad (\text{CF2})$$

aggregates those pieces without paying once per root. Coarea and (CF1) control the high pieces on average over the argument. Choose one argument realising that average, order the lift endpoints cyclically on ∂C_t , and use the shortest adjacent boundary arc. Averaging the two lifts and the boundary arcs over the k adjacent pairs gives (CF). The strict unit-sublevel statement chooses the regular level below the top of the window when $\lambda\mu \leq 1$. $\square$

The same estimate reaches the sharp target in explicit regions of the parent problem. Let k_0 be the number of roots in the component of K_μ containing the chosen minimising critical point. Let C be the corresponding component of $K_{2\mu}$ and put

$$\kappa = \frac{\text{cap}(\bar{C})}{(2\mu)^{1/n}}.$$

Thus $\kappa = e^{-\Sigma/n}$, where Σ is the total exterior Green defect of the roots excluded from C .

Theorem 2.2 (first-merge arity and capacity closures). *Assume all roots lie in the open unit disc and $\mu \leq 1/2$.*

- (i) *If $k_0 \geq 17$, two distinct roots are joined inside $\{|f| < 1\}$ by a path of length strictly less than 2.*
- (ii) *Put*

$$A = \frac{283}{3610}, \quad B = \frac{52029}{9100}, \quad \tau_k = \frac{\sqrt{2k} - A}{B}.$$

If $\kappa \leq \tau_{k_0}$, the same conclusion holds. In particular, $\kappa \leq 1/3$ suffices for every $k_0 \geq 2$, while at $k_0 = 16$ the rational cutoff improves to $\kappa \leq 39/40$.

The second part retains the component root count and replaces the global area bound in (CF1) by $\text{Area}(C) \leq \pi \text{cap}(\bar{C})^2$. Consequently a counterexample with $\mu \leq 1/2$ must have $k_0 \leq 16$ and must lie above its arity-dependent capacity cutoff. Smaller critical levels give stronger nested regions: $\mu \leq 1/4$ closes every $k_0 \geq 12$, $\mu \leq 1/8$ closes every $k_0 \geq 10$, and the $\mu \leq 1/64$ calculation raises the surviving $k_0 = 9$ capacity cutoff to $99/100$.

There is a complementary closure criterion that spends separation in the critical spectrum instead of component capacity. Suppose f is squarefree and a critical point c_* attaining μ is simple. Let $T_* > \mu$ be the supremum of the levels for which the component born at c_* contains exactly two roots. Component root counts change only at critical-value moduli, so T_* is at least the next distinct critical-value modulus.

Theorem 2.3 (minimal-hub spectral window). *Under these hypotheses, two roots are joined by a path inside K_μ of length at most*

$$\Psi_n(T_*/\mu) \mu^{1/n}, \quad \Psi_n(x) = x^{1/n} \sqrt{2 \log \frac{x+1}{x-1}}. \quad (\text{SW})$$

The function Ψ_n is strictly decreasing on $(1, \infty)$. Let x_n^ be the unique solution of $\Psi_n(x) = 2$ for $n \geq 3$. If the roots lie in a disk of radius R and the second distinct critical-value modulus v_2 satisfies $v_2/\mu > x_n^*$, then the path has length less than $2R$ and proves the parent conclusion. The first thresholds are*

$$\begin{aligned} x_3^* &= 1.6064943 \dots, & x_4^* &= 1.4784372 \dots, \\ x_5^* &= 1.4289343 \dots, & x_{10}^* &= 1.3598081 \dots \end{aligned}$$

and x_n^ decreases to 1.*

Indeed, the two-root descent-arc estimate at any $t \in (\mu, T_*)$ is

$$\text{len}(\Gamma) \leq t^{1/n} \sqrt{2 \log \frac{t + \mu}{t - \mu}}, \quad \Gamma \subset K_\mu.$$

Writing $x = \coth u$ gives $\Psi_n(x)^2 = 4u \coth(u)^{2/n}$ and proves strict decrease. Taking the limit $t \uparrow T_*$ gives (SW); the Fekete–Vandermonde bound $\mu \leq R^n$ converts $\Psi_n(\nu_2/\mu) < 2$ into the parent metric bound. The same criterion is stable under the compositional pullback $P((z - h)^q)$ after adjoining the hub value $|P(0)|$ to the critical spectrum. The complete proof and threshold certificates are in `ErdosProblems/Erdos1041/MinimalHubWindowJoin.md`. This theorem is silent at a multiple minimising critical point and makes no priority claim.

These statements are complete ordinary analytic proofs. They are not Lean theorems and make no literature-priority claim. Their full constant certificates, component-capacity tables, and local proofs are in `ErdosProblems/Erdos1041/UnconditionalConstantFactorBound.md`; the Pólya input is the global area–capacity inequality, not the local winding–coarea estimate, reciprocal-derivative aggregation, or assembled path theorem. The unrestricted constant 2 and the high-critical or low-arity/high-capacity residual cells remain open.

3 Critical proximity and straight-path obstructions

The strongest source-current theorem that applies in every degree is a sharp metric selection principle at a critical point. It consumes the complete logarithmic-derivative balance rather than a degree-specific coefficient pattern.

Theorem 3.1 (critical geometric-mean proximity). *Let $n \geq 2$, let $z_1, \dots, z_n, c \in \mathbb{C}$ with $c \neq z_k$ for every k , and suppose*

$$\sum_{k=1}^n \frac{1}{c - z_k} = 0.$$

If $r > 0$ is determined by

$$r^n = \prod_{k=1}^n |c - z_k|,$$

then there are distinct indices i, j such that

$$|c - z_i| + |c - z_j| \leq 2r.$$

The proof chooses the two smallest distances $d_i \leq d_j$. The reciprocal balance gives $d_j \leq (n - 1)d_i$, while minimality and the geometric-mean identity give $d_i d_j^{n-1} \leq r^n$. After normalising by r , a sharp two-variable Bernoulli inequality yields $d_i + d_j \leq 2r$. The complete complex theorem, including the selector and both inequalities, is checked as *exists two roots dist sum le two mul geom mean*.

This is not silently promoted to a path theorem. The same source module checks two exact barriers to the most tempting completions.

Proposition 3.2 (two straight-path no-go families). *There is a monic quintic with all roots in the open unit disc and a non-root critical point c whose unique nearest root has a point on the straight spoke to c outside $\{|f| < 1\}$. There is also a monic cubic with all roots in the open unit disc such that the midpoint of every pair of distinct roots lies outside $\{|f| < 1\}$.*

The certificates are exact rational-algebraic computations, checked as *nearest spoke unique nearest spoke escapes* and *all straight cubic every pair midpoint escapes*. The first defeats a tie-breaking escape hatch; the second defeats every straight root-pair segment. Together with Theorem 3.1, they isolate the real gap: critical-point proximity selects a short metric pair, but universal containment requires curved or topological geometry.

4 Solved polynomial families

Three source-current families do cross that containment gap. Their complete paper theorems are ordinary mathematics assembled from formalised load-bearing kernels; this section keeps that authority split explicit.

4.1 Translated cubic quotient fibres

Theorem 4.1 (translated cubic quotient fibres). *Let $q \geq 2$, $h \in \mathbb{C}$, and let P be a monic cubic. Put*

$$f(z) = P((z - h)^q).$$

If all zeros of f lie in the open unit disc and f has at least two distinct zeros, then two distinct zeros of f are joined by a two-segment path of length strictly less than 2 contained in $\{|f| < 1\}$.

For cubic roots r, s, v , the real charges $\Re(\overline{rs + v})$ sum to more than -3 , so one is greater than -1 . For that root, AM–GM and the exact cyclotomic cancellation $(1 - t)(1 + t + t^2) = 1 - t^3$ prove that its complete origin spoke is safe. Lean checks this fan-in as *cubic has safe root spoke*. Pulling the spoke back under $y \mapsto y^q$ gives q safe spokes from h ; two different fibre points are joined through h with length $2|y| < 2$. The finite root-of-unity mean-square argument and fibre assembly are ordinary, not hidden Lean claims.

4.2 Primitive sparse quintics

Theorem 4.2 (primitive sparse quintics). *Let*

$$p(z) = z^5 + az^4 + bz + c$$

be monic with all five root occurrences in the open unit disc. Then two distinct root occurrences w_i, w_j satisfy

$$|bw_i + c| < 1, \quad |bw_j + c| < 1,$$

and the corresponding two radial spokes join them through the origin inside $\{|p| < 1\}$ with total length strictly below 2. If the two occurrences coincide, the associated root-to-root path is degenerate.

The stronger closed-disc selector has non-strict tails; if $a \neq 0$ the selected tails are strict, while for $a = 0$ strictness is exactly the interior-root condition. The phase-sensitive

boundary theorem uses the first three Newton moments and a cubic separator. Its harmonic extension controls mixed interior/boundary configurations and shows that an unsafe interior tail contributes at most $2/31$. Lean checks the finite selectors *primitive boundary exists two tail energy lt one* and *primitive interior exists two tail energy lt one*; the rotation, moment identification, Abel path estimate, and two-spoke assembly are ordinary mathematics.

4.3 Sharp collinear roots

The complete all-degree collinear theorem appears in Section 12. It gives the sharp Chebyshev constant, not merely an existence bound, and therefore belongs to the solved-family layer rather than the frontier layer. Lean checks the constrained alternation engine *exists peak le of monic comparison* and its Chebyshev consumer *exists peak le comparison bound*; affine normalisation, root-gap selection, and transport back to the original line are ordinary.

5 Critical-value separation in every degree

The solved families of Section 4 are unconditional and complete, and each is confined to a shape of root configuration. This section gives a hypothesis that is not a shape at all. It constrains only the critical spectrum, it holds in every degree above an exact cutoff, and where it holds it gives the target constant 2 rather than a constant multiple of it.

Theorem 5.1 (critical-value separation). *Let f be monic of degree n , let c be a simple critical point with $v = f(c) \neq 0$, and let $S > 1$. Suppose every other critical point d of f satisfies*

$$\left|1 - \frac{f(d)}{v}\right| \geq S.$$

Then the square-root-resolved inverse branch through c joins two roots of f inside $\{|f| \leq |v|\}$ by a path of length at most

$$2|v|^{1/n}(1+S)^{1/n}\sqrt{\log \frac{S}{S-1}}.$$

In particular the connector is shorter than $2|v|^{1/n}$ whenever

$$(1+S)^{2/n} \log \frac{S}{S-1} < 1. \tag{1}$$

Corollary 5.2 (the exact convenient regimes). *Inequality (1) holds for $S = 4$ in every degree $n \geq 3$, for $S = 3$ in every degree $n \geq 4$, and for $S = 2$ in every degree $n \geq 6$. Each cutoff is the first degree at which its inequality holds: the squared coefficient of (1) is 1.43841 at $(n, S) = (2, 4)$, 1.02171 at $(3, 3)$, and 1.07566 at $(5, 2)$, while at $(6, 2)$ it is 0.99969.*

Corollary 5.3 (a parent-problem regime). *Let f be monic with all roots in the open unit disc, let c be a simple critical point with $v = f(c) \neq 0$, and suppose $|1 - f(d)/v| \geq 4$ for every other critical point d . If $n \geq 3$ then Erdős Problem #1041 holds for f .*

The separation hypothesis makes c the unique critical point of minimum modulus, the Fekete chain gives $|v| < 1$, and Theorem 5.1 with $S = 4$ places a connector of length below 2 inside $\{|f| < 1\}$.

The proof of Theorem 5.1 is the same area–capacity argument used in Section 2, run at a single saddle instead of over a merge tree. Normalise so that $c = 0$, $v = 1$, and $|f''(0)| \neq 0$. The square substitution resolves the saddle, so $f(Z(\zeta)) = 1 - \zeta^2$ has two local holomorphic solutions interchanged by $\zeta \mapsto -\zeta$. A finite branch point of the algebraic continuation of Z requires $f'(d) = 0$ and $\zeta^2 = 1 - f(d)$, and the separation hypothesis puts every such point on or outside $|\zeta| = \sqrt{S}$; properness of a polynomial rules out escape to infinity over a bounded value set, so the monodromy theorem continues Z holomorphically through $|\zeta| < \sqrt{S}$. It is injective there, because $Z(\zeta_1) = Z(\zeta_2)$ forces $\zeta_1^2 = \zeta_2^2$, and the two inverse sheets can meet only at an excluded branch point. Writing $Z(\zeta) = \sum_{k \geq 1} a_k \zeta^k$ and fixing $1 < R < \sqrt{S}$, injectivity and the area formula give

$$\pi \sum_{k \geq 1} k |a_k|^2 R^{2k} = \text{Area } Z(D_R),$$

while $f(Z(\zeta)) = 1 - \zeta^2$ confines the image to $\{|f| < 1 + R^2\}$, whose area is bounded by $\pi(1 + R^2)^{2/n}$ for a monic f . That bound is the composition of two published inequalities: Dubinin’s Theorem 1 [6, Thm. 1], which bounds the area of $\{|f| < s\}$ by $s^{2/n}$ times the area of $\{|f| < 1\}$, and Pólya’s classical inequality [7], which bounds the latter by π . Cauchy–Schwarz then gives

$$\sum_{k \geq 1} |a_k| \leq (1 + R^2)^{1/n} \left(\sum_{k \geq 1} \frac{1}{k R^{2k}} \right)^{1/2} = (1 + R^2)^{1/n} \sqrt{\log \frac{R^2}{R^2 - 1}},$$

and termwise integration of $|Z'|$ over $[-1, 1]$ bounds the connector length by $2 \sum_{k \geq 1} |a_k|$. Letting $R \uparrow \sqrt{S}$ proves the theorem.

Boundary. The analytic continuation, the univalence, the area formula, and the area–capacity inequality are ordinary mathematics; they are proved in `ErdosProblems/Erdos1041/FirstMergeCriticalValueSeparationCertificate.md` and are not formalised. What Lean checks is the numerical half of the statement, in `ErdosProblems/Erdos1041/FirstMergeCriticalValueSeparation.lean`: the squared coefficient *first merge squared coefficient* is antitone in the degree (*first merge squared coefficient \ antitone \ degree*), a squared connector length below 4 times a coefficient below 1 forces length below 2 (*first merge \ length \ lt \ two \ of \ squared \ bound*), and the three regimes of Corollary 5.2 hold exactly (*first merge \ exact \ convenient \ thresholds*). The Comparator entry is `ExternalVerification1041FirstMergeCriticalValueSeparation`. Lean does not prove the analytic hypothesis $\ell^2 \leq 4c(n, S)$ that those consumers take as input, and nothing in this section asserts the unrestricted problem or the complementary near-tie regime.

This regime is incomparable with Pendyala’s degree-four theorem [4, Thm. 1, p. 1], which is unconditional on the critical spectrum and confined to $n = 4$. Degree four with a near-tie in the critical spectrum is his and not ours; degree forty with $S = 4$ separation is ours and not his.

6 The Newton value equation

Away from the critical set, define the complex Newton field

$$N(z) = -\frac{f(z)}{f'(z)}.$$

Let $z(t)$ be differentiable with $z'(t) = N(z(t))$, and put $w(t) = f(z(t))$.

Theorem 6.1 (value equation). $w'(t) = -w(t)$.

The computation is one line: $w' = f'(z) z' = f'(z) \cdot (-f(z)/f'(z)) = -f(z) = -w$. The kernel checks it as *newton flow value has deriv at*, together with the differential form of the first integral, *newton flow scaled value has deriv at zero*:

$$\frac{d}{dt}(e^t f(z(t))) = 0, \quad \text{equivalently} \quad f(z(t)) = e^{-t} f(z(0)).$$

Observe what this says about the geometry. The value moves radially inward at exponential rate and never changes argument. The lemniscate $\{|f| < 1\}$ is therefore forward-invariant, and the flow lines are exactly the preimages of rays from the origin.

Corollary 6.2 (ray separation). *The endpoints of any finite Newton-flow connection lie on the same oriented ray from 0. Consequently, if two critical values lie on distinct positive rays, no Newton-flow saddle connection joins the corresponding critical points.*

This is checked in consumer form: the kernel accepts the implication from the hypothesis of distinct rays to the absence of a connection. It is the statement the topology needs, and it is a genuine sharpening of the criterion used in the literature.

7 Arguments, not moduli

It is tempting to arrange a generic perturbation so that the critical values are pairwise distinct, or that their moduli are pairwise distinct, and to conclude that saddle connections are excluded. Neither is enough.

Two distinct critical values can lie on one ray, and two critical values with distinct moduli certainly can: the ray records the argument, and the modulus is exactly the coordinate the flow contracts. By Corollary 6.2 the invariant that excludes connections is the argument. What a perturbation must therefore achieve is pairwise distinct critical-value *arguments*, which is a condition on $n - 1$ points modulo the circle rather than on their positions in the plane.

The cost of that condition is also checked, and it is small.

Theorem 7.1 (ray-collision locus). *Let $a \neq b$ be complex. Every common translation β for which $a + \beta$ and $b + \beta$ lie on the same positive ray has the form*

$$\beta = \frac{ra - b}{1 - r}, \quad r \in \mathbb{R}_{>0}, r \neq 1.$$

So each pair of critical values contributes a one-real-parameter forbidden locus in the translation plane, given in closed form (*translated same positive ray parameterization*). A finite union of such loci has empty interior, which is the shape one wants for an avoidance argument. Turning that into a perturbation of f requires root retention and length slack. Section 15 settles finite-exceptional coefficient genericity and quantitative root retention; global component stability and the transfer of strict metric slack remain open.

8 Exact reciprocal coordinates and zero-margin transfer

The local near-polygon analysis becomes substantially cleaner when its nonlinear terms are retained as coordinates. Assume throughout this section that

$$f(z) = \prod_{j=1}^n (z - a_j), \quad 0 < |a_j| < 1,$$

and put

$$r = \min_j |a_j|, \quad p_m = \sum_{j=1}^n a_j^{-m} \quad (m \geq 1).$$

Theorem 8.1 (exact reciprocal Newton expansion with C^2 tails). *For $|z| < r$,*

$$\log |f(z)| = \log |f(0)| - \Re \sum_{m \geq 1} \frac{p_m}{m} z^m, \quad (\text{R1})$$

and the series is absolutely convergent. Write $q = |z|/r < 1$ and

$$T_N(z) = \sum_{m > N} \frac{p_m}{m} z^m.$$

Then

$$|T_N(z)| \leq \frac{nq^{N+1}}{(N+1)(1-q)}, \quad (\text{R2})$$

$$|T'_N(z)| \leq \frac{n}{r} \frac{q^N}{1-q}, \quad (\text{R3})$$

$$|T''_N(z)| \leq \frac{n}{r^2} q^{N-1} \left(\frac{N}{1-q} + \frac{q}{(1-q)^2} \right). \quad (\text{R4})$$

The derivative bounds are read for $N \geq 1$; the value bound holds for $N \geq 0$.

Proof. Factorwise real logarithms avoid any global choice of a complex logarithm. The factorisation

$$\frac{f(z)}{f(0)} = \prod_j \left(1 - \frac{z}{a_j} \right)$$

and the absolutely convergent scalar series give

$$\log |f(z)| - \log |f(0)| = \sum_j \log \left| 1 - \frac{z}{a_j} \right| = -\Re \sum_j \sum_{m \geq 1} \frac{1}{m} \left(\frac{z}{a_j} \right)^m = -\Re \sum_{m \geq 1} \frac{p_m}{m} z^m.$$

This proves (R1). Since $|p_m| \leq nr^{-m}$, summing the geometric majorant gives (R2). Termwise differentiation and the identities for the first two differentiated geometric tails give (R3)–(R4). $\square$

The C^2 endpoint is load-bearing: moving quadratic contacts are invisible to a mere uniform-value estimate. Moreover, the coordinates p_m do not depend on a labelling choice and are rational in the polynomial coefficients.

Proposition 8.2 (coefficient rationality and the exact near-Fekete bridge). *Write $f(z) = \sum_{k=0}^n c_k z^k$, $c_n = 1$, $c_0 \neq 0$, and $\hat{c}_k = c_k/c_0$, extended by zero for $k > n$. Then $\hat{c}_0 = 1$ and, for every $m \geq 1$,*

$$p_m = -m\hat{c}_m - \sum_{j=1}^{m-1} \hat{c}_j p_{m-j}. \quad (\text{R5})$$

Thus every reciprocal coordinate is a rational function of the coefficients of f , with denominator a power of $f(0)$.

Now write a near-regular n -gon as

$$a_k = \omega^k(1 + e_k), \quad \omega = e^{2\pi i/n}, \quad \eta = \max_k |e_k| < 1,$$

and put $E_m^{(j)} = \sum_k e_k^j \omega^{-km}$ and $E_m = E_m^{(1)}$. For $1 \leq m \leq n-1$,

$$-\frac{p_m}{m} = E_m - \frac{m+1}{2} E_m^{(2)} + \frac{(m+1)(m+2)}{6} E_m^{(3)} - \dots, \quad (\text{R6})$$

whose j th term is

$$(-1)^{j-1} \frac{1}{m} \binom{m+j-1}{j} E_m^{(j)}.$$

In particular,

$$\left| -\frac{p_m}{m} - E_m \right| \leq \frac{n(m+1)\eta^2}{2(1-\eta)^{m+2}}. \quad (\text{R7})$$

Proof spine. Newton's identity for the roots a_j^{-1} of the reversed polynomial gives (R5) after using $e_k(a^{-1}) = (-1)^k \hat{c}_k$. For (R6), expand $(1 + e_k)^{-m}$ by the binomial series. The constant term cancels because $\sum_k \omega^{-km} = 0$ for $1 \leq m < n$; the remaining terms are exactly those displayed. Taylor's formula with integral remainder for $(1 + e)^{-m}$ gives (R7). $\square$

The conceptual point is stronger than the $O(\eta^2)$ estimate: the “anchored nonlinear remainder” is the second and higher part of the exact coefficient $-p_m/m$. It is not discarded error. This matters precisely at a tied Newton face, where a quadratic correction can have the same weight as a nominally active linear mode. The reciprocal chart retains both.

The coefficient-rational coordinates also give a compact local scale, but the normalisation is not itself rational or semialgebraic: it includes the real logarithm below and a local root branch. Near the regular n -gon, on a neighbourhood where $p_n \neq 0$, choose a continuous nonzero branch $\lambda^n = p_n/n$, put

$$b_m = (p_m/m)\lambda^{-m}, \quad d = -\log |f(0)|,$$

and set

$$\rho = \max \left\{ d^{1/n}, \max_{1 \leq m < n} |b_m|^{1/(n-m)} \right\}, \quad A_0 = d/\rho^n, \quad A_m = b_m/\rho^{n-m}.$$

Then $0 \leq A_0 \leq 1$, $|A_m| \leq 1$, and at least one coordinate lies on the boundary. After $z = \rho\lambda^{-1}\zeta$, retaining terms through any $N \geq n$ leaves, for each fixed M and all sufficiently small ρ with $\rho M/(r|\lambda|) < 1$, a C^2 error of order $O_{n,N,M}(\rho^{N+1-n})$ on $|\zeta| \leq M$. Hence the unmodelled tail can be pushed beyond any prescribed finite contact order. The chart does not itself prove that every limiting face transfers. In the present open-disc setting $d > 0$ and hence $\rho > 0$; the closure point $\rho = 0$, corresponding to the regular polygon after normalisation, is handled separately rather than by dividing by the scale.

Two exact elementary consequences close the local zero-margin bookkeeping.

Proposition 8.3 (staple identity and contact discriminant). *For distinct roots a_i, a_j , put $u_k = a_k/|a_k|$. If $0 \leq s \leq \min(|a_i|, |a_j|)$, the equal-radius staple*

$$[a_i, su_i] + [su_i, su_j] + [su_j, a_j]$$

has length

$$L = |a_i| + |a_j| - s(2 - |u_i - u_j|) \leq |a_i| + |a_j| < 2. \quad (\text{R8})$$

The strict inequality holds for every admissible s , including $s = 0$, whenever the roots lie in the open unit disc.

If $\kappa > 0$ and

$$F(x) \leq -\kappa x^2 + B|x| + V + \varepsilon,$$

then

$$F(x) \leq V + \varepsilon + \frac{\max\{B, 0\}^2}{4\kappa}. \quad (\text{R9})$$

Consequently the contact remains strictly negative whenever $V + \varepsilon + \max\{B, 0\}^2/(4\kappa) < 0$.

Proof. The two radial arms of the staple have lengths $|a_i| - s$ and $|a_j| - s$, while its chord has length $s|u_i - u_j|$, proving (R8); use $|u_i - u_j| \leq 2$. Completing the square in $-\kappa x^2 + B|x|$ gives the exact supremum $\max\{B, 0\}^2/(4\kappa)$ and proves (R9). When $B \geq 0$ this is the familiar $B^2/(4\kappa)$ formula. $\square$

Proposition 8.3 rules out two recurrent false moves: root-direction changes need not spend a polynomial-independent length margin, and uniform convergence to a nonpositive quadratic-contact model is not enough. The local transverse jet and curvature must satisfy the displayed sufficient negativity test. Because F is only bounded above by the quadratic majorant, this test is not asserted to be necessary for contact survival.

Theorem 8.1 and Propositions 8.2–8.3 are ordinary exact theorems. Lean checks the reciprocal power-sum majorant, the geometric tail, the solved Newton recursion, the staple equality and strict length consequence, and the completed-square bound and negativity consumer. It does not formalise the complex logarithmic series, the full C^2 tail, the near-Fekete bridge, the compact normalisation, or the general stratified transfer theorem. The classical expansion and Newton identity carry no novelty claim; the paper contribution here is the exact identification (R6) and the resulting coordinate choice at the transfer frontier.

9 A proof gap in the unrestricted argument

The March manuscript's Proposition 12 claims the following load-bearing statement. For $u = -\log |f|$, a connected component V of $\{u > c\}$ carrying $m \geq 2$ simple zeros, and the stated regularity and Morse hypotheses, it constructs, for every $\varepsilon > 0$, an embedded spanning tree $G_\varepsilon \subset V$ with

$$\text{len}(G_\varepsilon) \leq \frac{1}{2\pi} \int_{2\alpha}^{\infty} P_V(t) dt + \varepsilon. \quad (5.1)$$

The final theorem uses this estimate, so the issue below cannot be bypassed by calling the proposition auxiliary. Proposition 7 of the unattributed local revision repeats the same estimate.

The estimate itself is false. Take the Cassini polynomial $f_a(z) = z^2 - a^2$ at $a = 9/10$. Its component contains the two roots at distance $9/5$ and satisfies the relevant Morse and critical-value hypotheses. Direct level-length calculation gives the upper majorant

$$4(\sqrt{a^2 + a} - a) < \frac{41}{25}$$

for the right side of (5.1) before ε , whereas every connected set containing both roots has length at least $9/5$. Since $9/5 - 41/25 = 4/25$, choosing a smaller positive ε contradicts the assertion. Thus neither a different local saddle neighbourhood nor a perfect topological decomposition can recover the printed coefficient $1/(2\pi)$. The exact inequality and its abstract tree-budget contradiction are checked in `ErdosProblems.Erdos1041.CassiniTreeBudget`; the level-length majorant is an explicit analytic input, not a kernel-checked integral evaluation.

At an interior index-one critical point p , the proof invokes a Morse chart

$$u = \mu + x^2 - y^2$$

and replaces the saddle by a three-ended neighbourhood having one connected lower cross-section and two connected upper cross-sections. That local model is false as written. Because $p \in V$ and V is open, a sufficiently small closed disc around p lies entirely in V . In that disc the full Morse chart has four sectors: two components of $u > \mu$ and two components of $u < \mu$. A global component argument cannot delete one local sector from a disc already contained in V .

This independently diagnoses a proof step, but the Cassini witness above also refutes the proposition's statement. A different route might cut an adjoining regular annulus along a separatrix or regular flow arc before forming the block, retain a four-pronged saddle neighbourhood and change the assembly, or replace the local construction by the ray-cut decomposition proposed below. But no repair can retain (5.1); it must pay a positive attachment cost, select only one short pair instead of spanning every root, or use a different global metric inequality. The shorter descriptions of the same three-ended block do not repair the four-sector topology.

Corollary 6.2 supplies one independent input for a different route: distinct critical-value arguments exclude saddle-to-saddle Newton connections. It does not itself prove the compact planar decomposition, classify all orbit endpoints or provide the metric gluing estimate. Those are separate problems below.

10 Finite evidence

A search was run over random monic polynomials with roots in the unit disc. For each sample the region $\{|f| < 1\}$ was rasterised and shortest grid paths were computed between every pair of roots. The best upper bounds obtained were

degree 5,	500 trials :	1.1052648928
degree 6,	1500 trials :	0.8450414343
degree 8,	1500 trials :	0.6203916714
degree 10,	1500 trials :	0.4303640486.

No counterexample candidate was found, and the measured values sit well below the threshold 2. These are grid distances for the sampled configurations: upper bounds on those samples, not bounds over the family. The apparent decrease with degree is a property of the sample, and we draw no conjecture from it. Future searches should target named failure modes: near-degenerate saddles, thin necks, boundary-critical configurations and almost-connected separatrices—and report those diagnostics. Raster paths remain candidate finders, never continuous certificates.

11 Exact symmetric benchmark: complementary chords for every binomial degree

There is a complete all-degree model family which isolates the metric geometry without pretending to select paths for arbitrary polynomials. Let $f(z) = z^n - a$, where $n \geq 2$ and $0 < |a| < 1$. After rotation write $a = r^n > 0$, put $c = \cos(\pi/n)$, and set

$$r_* = (1 + c^n)^{-1/n}, \quad \varepsilon = (1 - r^n)^{1/n}.$$

Theorem 11.1 (complementary binomial chords). *Two adjacent zeros of $z^n - a$ can be joined by an explicit polygonal path inside $\{|z^n - a| < 1\}$ of length strictly below 2. For $r \leq r_*$ the adjacent-root chord itself works. For $r \geq r_*$, two radial legs and an inner adjacent crossing chord work after an arbitrarily small radial contraction. These two constructions meet at $r = r_*$.*

The central chord calculation is exact. On the chord between adjacent roots,

$$\max |z^n - r^n| = r^n(1 + c^n),$$

with the maximum at the midpoint; hence the first construction has precisely the threshold r_* . Above it, put $t = \varepsilon/c$. The path from r to t to $t\omega$ to $r\omega$ has its crossing chord in the closed lemniscate, and its length is

$$2r - 2\varepsilon \tan\left(\frac{\pi}{4} - \frac{\pi}{2n}\right) < 2r < 2.$$

The radius t is maximal: a larger crossing chord already escapes at its midpoint. Contracting that chord to radius λt with $0 < \lambda < 1$ makes the containment strict while keeping the two radial connections and the strict length bound. For $n = 2$, the adjacent-root diameter is already a strict path of length $2r < 2$.

The proof uses the sharp inequality

$$1 + \cos(n\theta) \leq 2 \cos^n \theta \quad (|\theta| \leq \pi/n),$$

followed by elementary chord geometry and the disk inequality that justifies the contraction. It is an authored ordinary all-degree proof with bounded numerical replay, not a Lean theorem or a priority claim. Its role is a symmetric benchmark: the exact transition says why a direct outer chord fails and why the complementary inner chord repairs it. It gives no path optimality, no theorem for arbitrary monic polynomials, and no solution of Erdős Problem 1.1.

12 The sharp collinear theorem: order buys an exact Chebyshev extremal

Collinearity supplies the global ordering that arbitrary complex root sets lack, and it yields a sharp all-degree solved subcase. Put

$$C_n = \frac{1}{2^{n-1} \cos^n(\pi/(2n))}.$$

Theorem 12.1 (sharp collinear root-diameter bound). *Let f be monic of degree $n \geq 2$, with collinear zero occurrences of diameter D . Then some two zero occurrences are joined by their straight segment, of length at most D , on which*

$$|f| \leq C_n \left(\frac{D}{2} \right)^n.$$

The constant is sharp: equality is attained by the affine images of the scaled Chebyshev root configuration with extreme roots at distance D .

Proof. Repeated zeros give the constant path, so suppose the zeros are distinct. Translation, rotation, and scaling by $R = D/2$ reduce to a monic real-rooted polynomial q with roots

$$-1 = y_1 < \dots < y_n = 1, \quad |f| = R^n |q| \text{ on the root line.}$$

Set $r_n = \cos(\pi/(2n))$ and compare q with the monic polynomial

$$q_*(x) = \frac{T_n(r_n x)}{2^{n-1} r_n^n}.$$

It vanishes at ± 1 and has $|q_*| \leq C_n$ on $[-1, 1]$. Choose a point c_i of maximum $|q|$ in each adjacent gap. If every gap maximum exceeded C_n , then $h = q - q_*$ would have the alternating signs of $q(c_i)$ at the c_i , hence a zero between each consecutive pair, as well as zeros at -1 and 1 . These are n distinct zeros, whereas the leading terms cancel and $\deg h \leq n - 1$, a contradiction. Thus one selected gap has $|q| \leq C_n$ throughout, and scaling back proves the bound.

For sharpness take the roots of q_* , namely

$$\frac{\cos((2k-1)\pi/(2n))}{r_n}, \quad 1 \leq k \leq n.$$

Their extreme roots are ± 1 , and every adjacent gap reaches C_n at a Chebyshev extremum. $\square$

If $D < 2$, the displayed level is below one and this segment lies in the open lemniscate. For $n \geq 3$, the same strict level holds already at $D = 2$; degree two is the exact boundary case. The proof is a complete authored ordinary argument. Two load-bearing alternation and Chebyshev-comparator kernels are Lean checked, while the affine normalization and transport remain ordinary; bounded replay is regression evidence only. This proves neither that every adjacent gap is safe nor any non-collinear analogue: the real order used in the alternation count is precisely what the general problem lacks.

13 An auxiliary critical-value theorem through degree five

A second route isolates the algebraic critical-value estimate from the metric path construction. For $c_1, \dots, c_m \in \overline{\mathbb{D}}$ put

$$A_j = \prod_{k=1}^m |1 - \overline{c_j} c_k|, \quad S_m = \sum_{j=1}^m A_j^{1/m}.$$

Write FP_m for the assertion $S_m \leq m$ for every such tuple.

Theorem 13.1 (four-point free-point inequality). FP_4 holds. More precisely,

$$\sum_{j=1}^4 \left(\prod_{k=1}^4 |1 - \overline{c_j} c_k| \right)^{1/4} \leq 4,$$

with equality if and only if $c_1 = c_2 = c_3 = c_4 = 0$.

The proof divides at $\max_j |c_j|^2 = 21/25$. In the central region, the logarithmic identities $\sum_j h_j = -E$ and $\sum_j h_j^2 \leq E$ mean L_j combine with an exact Bernstein-certified row envelope and rational exponential bounds. In the outer region, a K_4 matching Hölder inequality reduces the estimate to a one-variable radical inequality with positive exact squared slack. The two parts meet at the stated rational radius; neither is a numerical extrapolation from the other.

The central argument has an all-degree form that is stronger than the uniform radius estimate used above. Put

$$L_j = -\log(1 - |c_j|^2), \quad M_j = \frac{1}{m} \sum_{k=1}^m -\log(1 - |c_j| |c_k|),$$

and set $C(0) = 1/2$ and

$$C(t) = \frac{e^t - 1 - t}{t^2} \quad (t > 0).$$

Theorem 13.2 (adaptive all-degree central region). For every $m \geq 1$, the configuration-dependent condition

$$\frac{1}{m} \sum_{j=1}^m C(M_j) L_j \leq 1 \quad (\text{ROWCERT})$$

implies $S_m \leq m$, with equality only when all $c_j = 0$. In particular, let $L_* > 0$ be the unique positive solution of

$$e^{L_*} = 1 + 2L_*,$$

and put

$$\rho_* = (1 - e^{-L_*})^{1/2} = 0.8457729381 \dots$$

Then $S_m \leq m$ for every m whenever $\max_j |c_j| \leq \rho_*$.

Proof spine. Write $q_r = \sum_k c_k^r$ and

$$h_j = \log A_j^{1/m}, \quad E = \frac{1}{m} \sum_{r \geq 1} \frac{|q_r|^2}{r}.$$

The absolutely convergent logarithmic expansion gives

$$\sum_j h_j = -E, \quad h_j^2 \leq \frac{E}{m} L_j, \quad |h_j| \leq M_j.$$

On $[-M, M]$ the sharp quadratic remainder bound is

$$e^h \leq 1 + h + C(M)h^2.$$

Summing this row by row yields

$$S_m \leq m - E \left(1 - \frac{1}{m} \sum_j C(M_j) L_j \right) \leq m.$$

Equality forces every h_j and hence every power sum q_r to vanish; Newton's identities then force all $c_j = 0$. If $R = \max_j |c_j|$ and $L = -\log(1 - R^2)$, then $M_j, L_j \leq L$ and $C(L)L \leq 1$ is exactly $e^L \leq 1 + 2L$. This proves the uniform-radius consequence. $\square$

Thus any counterexample to FP_m must leave the explicit ball $\max_j |c_j| \leq \rho_*$ and must fail the sharper rowwise certificate. This is an unconditional infinite-degree region, but it is not a proof of FP_m without restrictions for $m \geq 5$.

The free-point statement enters the polynomial problem through a separate all-degree implication. Let $(S)_n$ denote the following assertion: if f is monic of degree n , the minimum enclosing radius of its roots is R , and $c_1, \dots, c_{n-1}$ are its critical points counted with multiplicity, then

$$\sum_{j=1}^{n-1} |f(c_j)|^{1/n} \leq (n-1)R.$$

Theorem 13.3 (free-point to critical values). *For every $n \geq 2$, FP_{n-1} implies $(S)_n$.*

Proof spine. After translating and scaling the minimum enclosing root disk, regard the left-hand side as a symmetric function of the roots in the closed unit polydisc. The critical values give a continuous plurisubharmonic multiset sum: away from critical-point collisions this follows from local holomorphic branches, and local boundedness gives

the extension across the collision locus. Applying the one-variable maximum principle in each root coordinate moves the maximum to the distinguished torus.

When all roots lie on the unit circle, the critical points satisfy the exact identity

$$|f(c_j)| = \prod_{k=1}^{n-1} |1 - \bar{c}_j c_k|.$$

Put $m = n - 1$ and $y_j = |f(c_j)|^{1/m}$. The hypothesis gives $\sum_j y_j \leq m$, while the tangent inequality

$$y^{m/(m+1)} \leq \frac{my + 1}{m + 1} \quad (y \geq 0)$$

converts the exponent and yields $\sum_j |f(c_j)|^{1/n} \leq m$. Scaling back restores the factor R . $\square$

The previously proved FP_3 case and Theorem 13.1 therefore establish $(S)_n$ for $n = 2, 3, 4, 5$; the new cases are degrees four and five. The proofs are ordinary exact analytic arguments with deterministic replay. Lean checks the scalar matching, stationary-point, and exponent-conversion kernels, but not the complete central/outer proof, plurisubharmonic removability, or the torus identity.

This advances the algebraic half of the programme but supplies neither a root-pair selection nor a path-length bound. In particular, $(S)_n$ does not imply Problem 1.1. The next free-point case is FP_5 .

14 The exact nonlinear lifetime currency

The merge-tree route has its own exact theorem. For each nontrivial child component u , let k_u be its number of roots and let $r_u = \beta_u / \beta_{\text{parent}(u)} \in (0, 1]$ be its merge-scale ratio. Its contribution to the attachment age of each descendant root is

$$x_u = \frac{1}{k_u} \log \frac{1}{r_u}.$$

The edge lifetime is written

$$\lambda_k(q) = \log \frac{1 + q^{2/k}}{1 - q^{2/k}}, \quad I_k(r) = \int_r^1 \frac{dq}{q \lambda_k(q)}.$$

Theorem 14.1 (exact attachment-age/lifetime currency). *Define*

$$\Phi(x) = \int_0^x \frac{dt}{\log(\coth t)} \quad (x \geq 0).$$

At $t = 0$ the integrand is understood by its continuous limiting value 0 (equivalently, the integral is improper at that endpoint). For every integer $k \geq 1$ and $0 < r \leq 1$, with $x = k^{-1} \log(1/r)$,

$$I_k(r) = k\Phi(x). \quad (\text{O1})$$

The function Φ is increasing and strictly convex on $(0, \infty)$, and

$$\frac{\Phi(x)}{x} \rightarrow 0 \quad (x \downarrow 0). \quad (\text{O2})$$

Consequently, for every fixed $k \geq 1$ and every $c > 0$, some $0 < r < 1$ satisfies

$$I_k(r) < c \frac{1}{k} \log \frac{1}{r}. \quad (\text{O3})$$

In particular, there is no positive universal linear conversion from attachment age to lifetime.

Proof. Set $q = e^{-kt}$. Then $dq/q = -k dt$, the endpoints $q = 1, r$ become $t = 0, x$, and

$$\frac{1 + e^{-2t}}{1 - e^{-2t}} = \coth t,$$

which proves (O1). Since $\coth t$, and hence $\log(\coth t)$, is strictly decreasing, the positive integrand $1/\log(\coth t)$ is strictly increasing. This proves monotonicity and strict convexity. Moreover

$$0 \leq \frac{\Phi(x)}{x} \leq \frac{1}{\log(\coth x)} \rightarrow 0,$$

which gives (O2); taking $r = e^{-kx}$ gives (O3). $\square$

The same monotonicity gives a useful non-linear lower bound that the no-linear floor does not erase. For $x > 0$,

$$\Phi(x) \geq \int_{x/2}^x \frac{dt}{\log(\coth(x/2))} = \frac{x}{2 \log(\coth(x/2))},$$

and therefore

$$I_k(r) \geq \frac{kx}{2 \log(\coth(x/2))}. \quad (\text{O3}')$$

The point is distribution-sensitive curvature, not a uniform positive linear coefficient at the origin.

This identity retains the complete distribution of edge ages. If a root chain has edges $u = 1, \dots, m$, set

$$A = \sum_u x_u, \quad K = \sum_u k_u, \quad L = \sum_u I_{k_u}(r_u).$$

Then the exact chain ledger and weighted Jensen inequality give

$$L = \sum_u k_u \Phi(x_u) \geq K \Phi\left(\frac{\sum_u k_u x_u}{K}\right) \geq K \Phi\left(\frac{2A}{K}\right), \quad (\text{O4})$$

where the last inequality uses $k_u \geq 2$ for ancestor edges. Separately, unweighted Jensen and the same inequality $k_u \geq 2$ give the cardinality form

$$L \geq 2 \sum_u \Phi(x_u) \geq 2m \Phi(A/m). \quad (\text{O4}')$$

It is not obtained by weakening the weighted K -bound: it records different data. For a strict component chain below an n -root top one also has the exact combinatorial ceilings

$$m \leq n - 2, \quad K \leq 2 + 3 + \dots + (n - 1) = \frac{n(n - 1)}{2} - 1.$$

These estimates are degree-dependent by design: splitting a fixed age among many very rapid mergers is exactly the regime in which a linear estimate loses all force.

The stronger global fact comes from summing across roots instead of following one root at a time. Fix an internal component v with root set S_v , and let $A_i(v)$ be the sum of the increments x_u along the path from root i to v . Let $u \leq v$ range over the nontrivial proper descendant components whose parent still lies below or equals v .

Theorem 14.2 (root-summed identity and two-young-root selector). *One has the exact identity*

$$\sum_{i \in S_v} A_i(v) = \sum_{u \leq v} \log \frac{1}{r_u} = \log \prod_{u \leq v} \frac{1}{r_u}. \quad (\text{O5})$$

If $|S_v| \geq 2$, there are distinct $i, j \in S_v$ such that

$$\max\{A_i(v), A_j(v)\} \leq \frac{\sum_{u \leq v} \log(1/r_u)}{|S_v| - 1}. \quad (\text{O6})$$

The constant $1/(|S_v| - 1)$ is sharp for the underlying ordered-age inequality.

Proof. The edge $u \rightarrow \text{parent}(u)$ contributes $k_u^{-1} \log(1/r_u)$ to exactly the k_u roots below it. Its coefficient after summing over roots is therefore one, proving the first equality in (O5); the second is the logarithm product law. Order the nonnegative ages as $a_1 \leq a_2 \leq \dots \leq a_{|S_v|}$. The largest $|S_v| - 1$ entries are at least a_2 , so $(|S_v| - 1)a_2 \leq \sum_i a_i$. Choosing the first two roots proves (O6). For arbitrary nonnegative age data the limiting vector $(0, c, \dots, c)$ shows sharpness of this ordered-age step. $\square$

The theorem is an unconditional attachment-compatible pair selector and an exact diagnosis of the metric frontier. It does *not* upper-bound the internal product $\prod_{u \leq v} 1/r_u$, show that the selected roots are compatible with one short embedded path, or perform the saddle and annular gluing. Those are the actual remaining producers. The source note proves the substitution, Jensen, and double-counting arguments and supplies bounded deterministic replay; Lean checks only the lifetime-to-Orlicz division, the weighted-chain order transfer, the abstract no-linear-floor implication, and the final second-age division.

15 Complements and further questions

The dependency chain has five separate gates. A proof, a minimally corrected hypothesis set, or an explicit polynomial or planar counterexample is a decisive answer to any one of them.

1. Repair or refute the saddle block

Problem 15.1 (corrected local saddle assembly). Under the hypotheses of Proposition 12, replace the invalid three-ended local model by a four-pronged block or by a block formed after a specified annular cut. Prove that for every $\eta > 0$ its connector has total Euclidean length below η , uniformly over the selected attachment points, and that the corrected blocks assemble to an embedded tree satisfying

$$\text{len}(G_\varepsilon) \leq \frac{1}{2\pi} \int_{2\alpha}^{\infty} P_V(t) dt + \varepsilon;$$

or give a polynomial or harmonic planar counterexample to that statement.

The Cassini family in §9 answers this problem negatively as stated: the requested global estimate is smaller than the distance between its two roots. What remains meaningful is a weaker one-pair assembly with a different metric budget; changing only the local saddle block cannot restore the coefficient in the displayed inequality.

The full-disc model $x^2 - y^2$, an annulus in which lower branches rejoin, two saddles joined by a separatrix and simultaneous saddle levels are mandatory tests. Repeating the one-lower/two-upper assertion does not answer the problem.

2. The compact ray-cut decomposition

Let p have simple roots, simple nonzero critical points and let $u = -\log |p|$. For regular values $c < T$, take

$$M_{c,T} = \overline{V \cap \{c < u < T\}},$$

and assume explicitly that this is a compact genus-zero surface, its lower boundary is one smooth Jordan curve, its upper boundary consists of m smooth root curves, all interior critical points are nondegenerate index-one saddles, the normalised gradient field

$$X = \frac{\nabla u}{|\nabla u|^2} = -\frac{p}{p'}$$

is transverse to the level boundaries, and no maximal X -trajectory has two saddle endpoints.

Problem 15.2 (finite strip decomposition after ray cuts). For every $\eta > 0$, construct disjoint Morse neighbourhoods N_j with $\sum_j \text{diam } N_j < \eta$ and a finite set of complete separatrix or regular-flow cuts so that every component of the complement is flow-diffeomorphic to a rectangle

$$[a_S, b_S] \times [0, 1], \quad u(\Phi_S(t, s)) = t,$$

with connected level sections and no uncut annular component. Give an explicit finite bound $E(s, m)$ for the number of strips, or exhibit the minimal missing hypothesis or a counterexample.

No particular formula such as $2s + 1$ is presumed. Boundary tangencies, simultaneous levels, branch reunion through an annulus and non-Hausdorff orbit spaces must be handled rather than suppressed.

3. Metric fan-in without losing the coefficient

For a strip S , write

$$\Gamma_t^S = S \cap \{u = t\}, \quad P_S(t) = \mathcal{H}^1(\Gamma_t^S),$$

and let k_S be its number of root ends. The flux identity

$$\int_{\Gamma_t^S} |\nabla u| ds = 2\pi k_S$$

gives the average trajectory estimate

$$\int_{\Gamma_{t_0}^S} \text{len}(\gamma_x) d\mu_{t_0}(x) = \frac{1}{2\pi k_S} \int_{a_S}^{b_S} P_S(t) dt.$$

Problem 15.3 (additive-error strip gluing). Assuming Problem 15.2, select trajectories in all strips and connect them through the saddle and root neighbourhoods so that, for every $\eta > 0$, the resulting embedded tree contains all m roots and obeys

$$\text{len}(G) \leq \frac{1}{2\pi} \int_{2\alpha}^{\infty} P_V(t) dt + \eta. \quad (6.1)$$

The total saddle, annular-cut and root-cap cost must be below η without a multiplicative loss in $1/(2\pi)$.

For the final strict inequality, use the actual collar slack

$$q = \frac{1}{2\pi} \int_{\alpha}^{2\alpha} P_V(t) dt > 0$$

and give budgets that keep the perturbation, tree error and transfer cost below fixed fractions of q . Independently choosing a shortest trajectory in each strip is not enough. Theorems 14.1 and 14.2 identify the exact missing control: a linear lifetime charge is impossible, while an upper bound on one internal merge-ratio product would select two young roots before the attachment age fragments across many edges.

4. Coefficient perturbation and stability

The constant-translation stage is no longer open. Once a finite critical-value family is injective, Lean proves an arbitrarily small translation making every value nonzero and pairwise positive-ray separated (*exists small translation separating arguments*). It also proves the explicit root-retention estimate; a shift below ε keeps all roots in the unit disc when

$$((n+1)\varepsilon)^{1/n} + \rho < 1$$

(*constant perturbation roots in unit disk*). A constant translation cannot separate initially equal critical values.

Proposition 15.4 (a one-parameter critical-value genericity lemma). *Let $f \in \mathbb{C}[z]$ have degree at least 2. There is a finite set $E \subset \mathbb{C}$ such that, for every $\lambda \notin E$, the polynomial $f + \lambda z$ has only simple critical points and its critical values are pairwise distinct. In particular, such λ occur arbitrarily close to 0.*

Proof. Write $d = \deg f$ and let a be the leading coefficient. A multiple critical point of $f + \lambda z$ satisfies

$$f'(u) + \lambda = 0, \quad f''(u) = 0.$$

Since f'' is not the zero polynomial, this excludes only finitely many values $\lambda = -f'(u)$.

For distinct critical points u, v with equal critical values, eliminating λ gives the two polynomial equations

$$A(u, v) := \frac{f'(u) - f'(v)}{u - v} = 0, \quad B(u, v) := \frac{f(u) - f(v)}{u - v} - f'(u) = 0, \quad (4.1)$$

and then $\lambda = -f'(u)$. These two equations have only finitely many solutions. Indeed, their top homogeneous parts are

$$A_{d-2} = da(u^{d-2} + u^{d-3}v + \dots + v^{d-2}),$$

and

$$B_{d-1} = a(u^{d-1} + u^{d-2}v + \dots + v^{d-1} - du^{d-1}).$$

They have no common projective zero at infinity. For $u \neq 0$, putting $r = v/u$, the first equation gives $1 + r + \dots + r^{d-2} = 0$, hence $r^{d-1} = 1$ and $r \neq 1$; the second expression, divided by au^{d-1} , is then $1 - d \neq 0$. If $u = 0$, the first expression is nonzero (and for $d = 2$ it is already a nonzero constant). Thus A and B have no common curve component: any such affine component would have a point at infinity. Their common zero set is finite, so the associated values $-f'(u)$ form a finite exceptional set. Avoiding its union with the first finite set proves the claim. $\square$

Proposition 15.5 (compact barrier stability). *Let $C, H \subset \mathbb{C}$ be compact, let $D_1, \dots, D_m$ be pairwise disjoint closed discs, and let $T \subset \mathbb{R}_{>0}$ be finite. Suppose that C and H are nonempty, that $m \geq 1$, that T is nonempty, that f' has no zero on C , that f has no zero on any ∂D_i , and that $|f(z)| \neq t$ for every $z \in H$ and $t \in T$. Put*

$$c_C = \min_C |f'|, \quad c_D = \min_i \min_{\partial D_i} |f|, \quad c_H = \min_{t \in T} \min_H ||f| - t|,$$

and

$$R = \max\{|z| : z \in H \cup \partial D_1 \cup \dots \cup \partial D_m\}.$$

Then $c_C, c_D, c_H > 0$. If

$$|\lambda| < c_C, \quad R|\lambda| + |\beta| < \min(c_D, c_H), \quad (4.2)$$

then $g_{\lambda, \beta} = f + \lambda z + \beta$ has no critical point in C , has the same number of zeros as f in every D_i (with multiplicity), and lies on the same side of every level $t \in T$ as f at every point of H .

Proof. Compactness and the hypotheses give the three positive minima. Since $g'_{\lambda, \beta} = f' + \lambda$, the first inequality in (4.2) gives

$$|g'_{\lambda, \beta}(z)| \geq |f'(z)| - |\lambda| > 0 \quad (z \in C).$$

On every ∂D_i the second inequality gives

$$|\lambda z + \beta| \leq R|\lambda| + |\beta| < |f(z)|,$$

so Rouché's theorem preserves the zero count in D_i . Finally,

$$\|g_{\lambda,\beta}(z) - f(z)\| \leq |\lambda z + \beta| < \|f(z) - t\| \quad (z \in H, t \in T),$$

which preserves the sign of $|f(z) - t|$. $\square$

This compact estimate separates several local requirements that were previously bundled into “stability”: protected collars and truncation boundaries remain free of critical points, prescribed root discs retain their zeros, and finite level barriers retain their chosen side. It does not assert that a selected sublevel component remains connected or nested, and it gives no continuity statement for the coarea slack q . The first, protected-set noncriticality implication is separately Lean-checked in `noncritical_on_of_norm_lt_uniform_lower_bound` and is targeted by the external Comparator interface. The Rouché root-count and level-side implications in this proposition remain ordinary paper proofs.

Problem 15.6 (two-stage generic perturbation with remaining global stability). For fixed root discs, compact collar K , regular levels $\alpha/2, \alpha, 2\alpha, 5\alpha/2$ and collar slack $q > 0$, prove that an arbitrarily small

$$g_{\lambda,\beta}(z) = f(z) + \lambda z + \beta$$

can be chosen so that:

1. $f + \lambda z$ has simple relevant critical points and injective complex critical values;
2. the checked constant translation β makes those values nonzero and pairwise ray-separated;
3. the relevant component remains in K , contains exactly the corresponding perturbed roots and has slack $q_g > q/2$; and
4. root displacement and straight-line transfer back to the original roots consume less than a prescribed fraction of q/m .

Proposition 15.4 proves the first item, with an arbitrarily small one-coefficient perturbation. Proposition 15.5 supplies the local collar, truncation, root-disc and finite-level controls once their compact margins have been fixed. The remaining issue is global component and strict-slack stability, followed by the quantified root-transfer budget.

Finite planar avoidance and the subsequent constant translation must not be relisted as missing. Critical-value genericity and compact barrier control are now elementary inputs; global component stability, strict slack stability and the transfer budget remain the open content.

5. The global Newton-flow claim ceiling

What the constant-factor theorem leaves open. Theorems 2.1 and 2.2 remove the former absence of any unconditional global path bound. They also close substantial parent regimes. They do not prove the sharp constant 2 on the remaining low-arity, high-component-capacity cells or on the high-critical branch. The Newton-flow programme below is therefore no longer needed to obtain a finite global constant; its purpose is to recover sharp containment and sharp metric allocation on precisely those residual cells. Pólya's primary theorem supplies only the global area–capacity input used in Section 2; no priority claim is inferred for the source-specific local estimates or assembled paths.

Problem 15.7 (relative global Newton-flow theorem). On a compact regular band

$$M_{a,b} = \{z : a \leq -\log |p(z)| \leq b\},$$

prove real-time existence and the identities

$$p(z(t)) = e^{-(t-t_0)} p(z(t_0)), \quad u(z(t)) = u(z(t_0)) + t - t_0$$

throughout every maximal orbit; classify both limiting endpoints among the regular boundary, root ends and finite saddle set; and exclude recurrent, accumulating and non-Hausdorff orbit phenomena. Under pairwise ray separation of the critical values, decide whether the flow is Morse–Smale relative to the boundary or whether the strongest valid conclusion is only absence of saddle-to-saddle connections.

The checked algebra proves the pointwise value equation and the endpoint-ray consumer. It does not supply global solution theory or an orbit-space graph. A positive stronger theorem must give the graph and a finite edge bound; a negative answer should exhibit the simplest ray-separated polynomial carrying the remaining pathology.

The auxiliary free-point route now has a different boundary. Theorem 13.1 closes FP_4 , Theorem 13.2 proves an explicit all-degree central region and a sharper adaptive certificate, and Theorem 13.3 transfers FP_{n-1} to $(S)_n$ in every degree. What remains there is unrestricted FP_m for $m \geq 5$ and, independently, a metric or selection theorem that converts critical-value control into the short root-to-root path.

Erdős #1041 remains open. The source now publicly verifies the Newton kernel, finite ray avoidance, quantitative constant-translation root control, and the protected-set noncriticality inequality in the compact-barrier argument; ordinary proofs supply finite-exceptional coefficient genericity and the remaining root-disc and level-barrier controls. Global component stability, the corrected planar decomposition and metric gluing remain the exact unresolved producers.

16 Evidence and boundary ledger

This ledger separates proved statements, checked kernels, computations, and open steps. Each entry states both the evidence and the limit of the claim.

Result or mechanism	Evidence and exact boundary
Principal results	
Critical two-root proximity	LEAN CHECKED. At every non-root critical point, two distinct roots have total critical-point distance at most twice the geometric mean of all root distances. The theorem selects a metric pair; it does not prove that a short connecting path stays in the lemniscate.
Straight-path obstructions	LEAN-CHECKED COUNTEREXAMPLES. Exact families rule out both the unique-nearest spoke and every pair of straight spokes through the critical point as universal constructions.
Translated cubic quotient fibres	CHECKED CUBIC KERNEL; ORDINARY FINITE-FIBRE ASSEMBLY. This gives a complete solved family $f(z) = P((z - h)^q)$. Lean proves the safe cubic spoke; the root-of-unity assembly is an ordinary proof.
Primitive sparse quintics	CHECKED SELECTOR KERNELS; ORDINARY PATH ASSEMBLY. Two tail-energy selectors give the solved open-disc family. The moment-to-path transport is an ordinary proof.
Sharp collinear family	CHECKED KERNELS; ORDINARY AFFINE TRANSPORT. This is a complete all-degree family with the sharp Chebyshev constant.
Critical-value and free-point route	
Newton identities	LEAN CHECKED. Away from critical points, a trajectory tangent to $-f/f'$ satisfies $w' = -w$ and $\frac{d}{dt}(e^t f(z(t))) = 0$.
Critical-value rays	LEAN CHECKED, CONSUMER FORM. Endpoints of a finite connection share one oriented ray, so distinct rays exclude a connection. For a pair of critical values, collision occurs on $\beta = (ra - b)/(1 - r)$ with $r > 0$ and $r \neq 1$.
Quartic case	CITED THEOREM. The degree-four result is proved in [4, Thm. 1, p. 1]. It does not extend to arbitrary degree.
FP_4	ORDINARY EXACT PROOF; DETERMINISTIC REPLAY. Four free points in $\overline{\mathbb{D}}$ satisfy the product-distance mean inequality, with equality only at the all-zero tuple. Lean checks scalar certificate kernels, not the full analytic proof.
All-degree central region	ORDINARY EXACT PROOF; DETERMINISTIC REPLAY. The adaptive rowwise certificate proves the free-point inequality in every degree; uniformly it covers $\max_j c_j \leq 0.8457729381 \dots$. This restricts every possible higher-degree counterexample but does not prove unrestricted FP_m for $m \geq 5$.
Continued on the next page	

Result or mechanism	Evidence and exact boundary
$FP_{n-1} \Rightarrow (S)_n$	ORDINARY EXACT PROOF; DETERMINISTIC REPLAY. The proof uses a plurisubharmonic reduction to the root torus, the torus product identity, and an exact exponent conversion. The several-complex-variables step is not formalised in Lean.
Critical-value inequality $(S)_n$	PROVED FOR $2 \leq n \leq 5$. Degrees four and five are new here. This auxiliary inequality alone does not produce a root-to-root path.

Attachment-time and local-transfer route

Attachment-age currency	ORDINARY EXACT PROOF; CHECKED SCALAR CONSUMERS. $I_k(r) = k\Phi(k^{-1} \log(1/r))$, and weighted Jensen bounds hold. No positive uniform linear age charge can hold.
Two-young-root selector	ORDINARY EXACT PROOF; FINITE REGRESSION. Subtree-size denominators cancel exactly, and two roots meet the sharp total-age threshold. The result does not supply a merge-product or geometric compatibility bound.
Reciprocal Newton expansion	ORDINARY EXACT PROOF; CHECKED FINITE KERNELS. The expansion has coefficient-rational Taylor coordinates and explicit C^0 , C^1 , and C^2 tails. It retains the anchored nonlinear remainder coefficientwise.
Staple and quadratic contact	ORDINARY EXACT PROOF; CHECKED SCALAR KERNELS. The staple spends no length margin. Completing the square gives a sharp quadratic-majorant bound and a sufficient negativity criterion for the underlying contact. A global stratified transfer theorem remains open.
Global constant-factor path	ORDINARY EXACT PROOF; FINITE REGRESSION ONLY. Two zero occurrences admit a path of length at most $(71/10)\mu^{1/n}$ in $K_{2\mu}$; when $\mu \leq 1/2$, the length is at most 5.7 in $\{ f < 1\}$. This result has no Lean or independent-review claim.
First-merge parent regimes	ORDINARY EXACT PROOF. The target constant 2 holds at $\mu \leq 1/2$ for first-merge arity at least 17, and under the explicit arity-dependent component-capacity cutoffs of Theorem 2.2. The complementary cells and the high-critical branch remain open.

Genericity, failed mechanisms, and open steps

March 2026 proof mechanism	REFUTED. A Cassini witness refutes the tree budget in [3, Prop. 12]; the local saddle block also fails independently.
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Continued on the next page

Result or mechanism	Evidence and exact boundary
Constant translation	LEAN CHECKED. Critical-value injectivity permits arbitrarily small ray avoidance while retaining an explicit unit-disc root margin.
Coefficient perturbation	PARTLY FORMALISED. The note proves finite-exceptional genericity; Lean checks protected-set noncriticality. Root-disc and level-barrier control remain ordinary arguments, while global component and strict-slack stability remain open.
Reeb decomposition and length fan-in	OPEN. These are the two remaining producers in the Newton-flow route.
Random search through degree 10	VERIFIED FINITE INSTANCES. The computations give upper bounds only for the sampled configurations.
Erdős #1041	OPEN. No proof of the general conjecture is claimed.

Statements and declarations

This manuscript is authored exposition, not Lean proof authority. Its strongest unrestricted-geometry paper theorem is the ordinary analytic constant-factor path theorem, together with the first-merge arity and component-capacity regimes that reach the parent constant 2. Lean does not formalise their coarea, univalent-map, component-capacity, or path-assembly arguments.

Its strongest source-current checked family is the critical geometric-mean proximity theorem *exists two roots dist sum le two mul geom mean*, paired with the exact unique-nearest-spoke and all-straight-cubic no-go certificates. The translated cubic quotient-fibre theorem, primitive sparse-quintic theorem, and sharp collinear theorem are complete ordinary paper theorems whose load-bearing selectors or alternation kernels are Lean checked. Lean does not formalise their finite-fibre, moment-to-path, or affine-transport assemblies.

The older checked core is the Newton value equation, the exponential first integral, the consumer form of ray separation, the finite planar-avoidance theorem, quantitative constant-translation root retention, and the ray-collision parameterisation. The decomposition and length statements of §15 are not proved. The diagnosis of Proposition 12 in §9 has two independent parts: the printed local saddle construction is topologically invalid, and the exact Cassini budget refutes the proposition’s metric statement. The search results of §10 are computations.

The four-point free-point inequality, the adaptive all-degree central-region theorem, and the all-degree $\text{FP}_{n-1} \Rightarrow (S)_n$ bridge are ordinary exact analytic theorems with green deterministic replay. The files `FreePointFP4Complete.lean` and `FreePointTorusPshReduction.lean` check scalar algebraic kernels only. They do not formalise the complete central/outer analysis, plurisubharmonic removability, the torus product identity, or a root-to-root path. No novelty or literature-priority claim is made for these auxiliary inequalities.

The exact attachment-age/lifetime transform, its Jensen consequences, the root-summed age/product identity, and the two-young-root selector are ordinary exact theorems proved in the source note `AttachmentAgeLifetimeOrlicz.md`. The companion Lean module checks only four scalar consumers after the analytic identities and monotonicity have been supplied. The five Comparator packages expose those four consumer types: `OrliczNoUniformLinear` and `NoUniformLinear` are duplicate interfaces for one mathematical obstruction, not two distinct paper results. No claim is made here that Lean formalises the integral substitution, convexity, Jensen’s inequality, the merge-tree double count, or the remaining geometric allocation theorem.

The reciprocal Newton expansion, coefficient recursion, near-Fekete bridge, compact normalisation, staple identity, and completed-square contact criterion are ordinary exact theorems proved in `ReciprocalNewtonExpansion.md`. Lean checks seven distinct finite kernel types exposed through nine Comparator packages: `ReciprocalTail` duplicates `NormRecipPowerSum`, and `ContactAbsorption` duplicates `ContactSurvives`. These are one reciprocal-coordinate and transfer family, not nine independent paper claims. The logarithmic-series identity, full C^2 bounds, exact near-Fekete identification, and rational compactification remain ordinary proof rather than Lean-formalised assertions.

A Guide to the formal sources

The source-current checked summit is distributed across `CriticalTwoRootProximity`, `CubicQuotientFiberCase`, `PrimitiveQuinticBoundaryTail`, `PrimitiveQuinticInteriorTail`, `SharpCollinearAlternation`, and `SharpCollinearChebyshev`. The public `ErdosProblems.Erdos1041.NewtonFlowRaySeparation` module contains the older checked Newton-flow source. The search of §10 is `scripts/search_counterexample.py` in the source package. The declaration table below is pinned to the shared formal-source commit used for the Newton-flow declarations in this problem-note series.

- [exists two roots dist sum le two mul geom mean](#)
- [nearest spoke unique nearest spoke escapes](#)
- [all straight cubic every pair midpoint escapes](#)
- [cubic has safe root spoke](#)
- [primitive boundary exists two tail energy lt one](#)
- [primitive interior exists two tail energy lt one](#)
- [exists peak le of monic comparison](#)
- [exists peak le comparison bound](#)
- [newton flow vector](#)
- [derivative mul newton flow vector](#)
- [newton flow value has deriv at](#)
- [newton flow scaled value has deriv at zero](#)

- same positive ray
- same positive ray refl
- same positive ray symm
- same positive ray trans
- translated same positive ray parameterization
- real affine line
- real affine line eq affine span
- is closed real affine line
- dense compl real affine line
- exists small avoiding finite real affine lines
- same positive ray imp mem real affine line
- exists small translation separating arguments
- norm lt one of near root
- constant perturbation root near original
- constant perturbation roots in unit disk
- same positive ray of real exp decay
- no newton connection of not same positive ray

Source-current auxiliary proofs. The proofs of Theorems 13.1 and 13.2 are ErdosProblems/Erdos1041/FreePointZeroInsertionFP4Matching.md; its deterministic replay is scripts/check_erdos1041_free_point_zero_insertion_fp4.py. The scalar Lean companion contains three_mul_pairProduct_le_sum_sq and the five outer_* stationary-point certificates. The proof of Theorem 13.3 is ErdosProblems/Erdos1041/FreePointTorusPshReduction.md; its replay is scripts/check_erdos1041_free_point_torus_psh_reduction.py. Its Lean companion checks rpow_le_tangent_at_one, sum_rpow_le_card_of_sum_le_card, and the two finite exponent bridge declarations. These source-current files are not covered by the older Newton-flow commit pin below; the paper states their separate authority and validation boundary explicitly.

The attachment-age/lifetime proof is ErdosProblems/Erdos1041/AttachmentAgeLifetimeOrlicz.md, and its scalar Lean companion is ErdosProblems/Erdos1041/AttachmentAgeLifetimeOrlicz.lean. Its Comparator interfaces are ExternalVerification1041LifetimeOrliczBound, ExternalVerification1041WeightedChainConsumer, ExternalVerification1041SecondAgeBound, and ExternalVerification1041OrliczNoUniformLinear. These interfaces cover the formalised consumers, not the ordinary analytic core of Theorems 14.1–14.2.

The reciprocal-coordinate proof and replay description are `ErdosProblems/Erdos1041/ReciprocalNewtonExpansion.md`; its finite Lean kernel is `ErdosProblems/Erdos1041/ReciprocalNewtonExpansion.lean`, and its exact replay is `ErdosProblems/Erdos1041/scripts/check_erdos1041_reciprocal_newton_expansion.py`. The Comparator interfaces are `ExternalVerification1041NewtonSolve`, `ExternalVerification1041NormRecipPowerSum`, `ExternalVerification1041GeometricTail`, `ExternalVerification1041QuadraticContact`, `ExternalVerification1041ContactSurvives`, `ExternalVerification1041StapleEquality`, and `ExternalVerification1041StapleLength`. They cover the seven checked finite kernels identified above, not every ordinary theorem in Section 8.

The finite near-Fekete result described in the abstract is the ordinary analytic source note `ErdosProblems/Erdos1041/NearFeketeTransverseClosure.md`, pinned to source commit `0f248d8826eca5f946052c1d4fd2f983bc3d7a04` and SHA-256 `3fbfcab1310c5d596990baab102d5787331f1df37a85b8becb2aab77eb5daadd`. Its limited formal consumer is `ErdosProblems.Erdos1041.NearFeketeTransverseClosure`, whose tracked Lean source is pinned to commit `ee2bfdebfe6d7d4827cf0d1b243f37fd3667c3e7` and SHA-256 `20cd39f660e19c71bb47868721e7f69b6ae5585045087237b0bd6a9fa64d43a5`. The declarations at lines 19–60 check only eventual sign and length transfer; they do not formalize the two-cutoff complex analysis, tangent strata, bulk regime, or the unrestricted problem.

Immutable source-generation boundary. The declaration table above is pinned to formal-source commit `7e80158d80c9becbbaa930a0b45da2db70f253d4` (23 August 2026), whose tracked source file `ErdosProblems/Erdos1041/NewtonFlowRaySeparation.lean` has SHA-256 `b3cde5cb40fb93fbbd410e8931cd9ae6fcb4e30e1346ac62d62ebae55f65fb1`. This pin covers only the declarations named above, including the exact value-equation, ray-separation, finite-avoidance, and root-retention locators; it is not a claim that the surrounding global-flow or length argument is formalised. The bounded project entrypoint for replay is `formal_math/erdos257_period_noncollapse/scripts/lean_fast_build.py` with the module target `ErdosProblems.Erdos1041.NewtonFlowRaySeparation`; a replay of that target remains validation evidence for the listed declarations, not an independent proof of the paper’s ordinary arguments.

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